

# **Modern academy**

## **Project of Numerical Analysis**

**Class:C12**

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# **Seidel Iterative Methods**

## **The Jacobi Method**

**The system given by :**

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots a_{2n}x_n = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + \dots a_{3n}x_n = b_3$$

.  
.  
.

$$a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \dots a_{nn}x_n = b_n$$

**Main idea of Jacobi :**

**To begin, solve the 1<sup>st</sup> equation for  $x_1$  the 2<sup>nd</sup> equation for  $x_2$  the 3<sup>rd</sup> equation for  $x_3$  so on to obtain the rewritten equations:**

$$x_1^{m+1} = \frac{1}{a_{11}} (b_1 - a_{12}x_2^m - a_{13}x_3^m - \dots a_{1n}x_n^m)$$

$$x_2^{m+1} = \frac{1}{a_{22}} (b_2 - a_{21}x_1^m - a_{23}x_3^m - \dots a_{2n}x_n^m)$$

$$x_3^{m+1} = \frac{1}{a_{33}} (b_3 - a_{31}x_1^m - a_{32}x_2^m - \dots a_{3n}x_n^m)$$

.

.

.

$$x_n^{m+1} = \frac{1}{a_{nn}} (b_n - a_{n1}x_1^m - a_{n2}x_2^m - \dots a_{n,n-1}x_{n-1}^m)$$

**Then make an initial guess of the  $x^{(0)} = (x_1^{(0)}, x_2^{(0)}, x_3^{(0)}, \dots x_n^{(0)})$  Substitute these values into the right hand side of the rewritten equations to obtain the first approximation,  $(x_1^{(1)}, x_2^{(1)}, x_3^{(1)}, \dots x_n^{(1)})$**

**This accomplishes one iteration.**

**In the same way, the second approximation  $(x_1^{(2)}, x_2^{(2)}, x_3^{(2)}, \dots, x_n^{(2)})$  is computed by substituting the first approximation's values into the right hand side of the rewritten equations.**

**By repeated iterations, we form a sequence of approximations  $x^{(m)} = (x_1^{(m)}, x_2^{(m)}, x_3^{(m)}, \dots, x_n^{(m)})$   $m = 1, 2, 3, \dots$**

**Example. Apply the Jacobi method to solve :**

$$8x_1 + 2x_2 - 3x_3 = 21$$

$$-x_1 - 10x_2 + 4x_3 = 13$$

$$2x_1 + 3x_2 + 9x_3 = -15$$

***Simplifying three equations :***

$$x_1^{m+1} = \frac{1}{a_{11}} (b_1 - a_{12}x_2^m - a_{13}x_3^m - \dots a_{1n}x_n^m)$$

$$x_1^{m+1} = \frac{1}{8} (21 - 2x_2^m + 3x_3^m)$$

$$x_2^{m+1} = \frac{1}{a_{22}} (b_2 - a_{21}x_1^m - a_{23}x_3^m - \dots a_{2n}x_n^m)$$

$$x_2^{m+1} = \frac{1}{-10} (13 + x_1^m - 4x_3^m)$$

$$x_3^{m+1} = \frac{1}{a_{33}} (b_3 - a_{31}x_1^m - a_{32}x_2^m - \dots a_{3n}x_n^m)$$

$$x_3^{m+1} = \frac{1}{9} (-15 - 2x_1^m - 3x_2^m)$$

$$(x_1^{(0)}, x_2^{(0)}, x_3^{(0)}, \dots x_n^{(0)}) = 0$$

**Let m = 0 (1<sup>st</sup> iteration) :**

$$x_1^{0+1} = \frac{1}{a_{11}} (b_1 - a_{12}x_2^0 - a_{13}x_3^0)$$

$$x_1^{0+1} = \frac{1}{8} (21 - 2x_2^0 + 3x_3^0)$$

$$x_1^1 = \frac{1}{8} (21)$$

$$x_1^1 = \frac{21}{8}$$

$$x_2^{0+1} = \frac{1}{a_{22}} (b_2 - a_{21}x_1^0 - a_{23}x_3^0)$$

$$x_2^{0+1} = \frac{1}{-10} (13 + x_1^0 - 4x_3^0)$$

$$x_2^1 = \frac{1}{-10} (13)$$

$$x_2^1 = -1.3$$

$$x_3^{0+1} = \frac{1}{a_{33}} (b_3 - a_{31}x_1^0 - a_{32}x_2^0)$$

$$x_3^{0+1} = \frac{1}{9} (-15 - 2x_1^0 - 3x_2^0)$$

$$x_3^{0+1} = \frac{1}{9} (-15 - 2x_1^0 - 3x_2^0)$$

$$x_3^1 = \frac{1}{9} (-15)$$

$$x_3^1 = \frac{-5}{3}$$

**m = 1 (2<sup>nd</sup> iteration) :**

$$x_1^{1+1} = \frac{1}{a_{11}} (b_1 - a_{12}x_2^1 - a_{13}x_3^1)$$

$$x_1^{1+1} = \frac{1}{8} (21 - 2x_2^1 + 3x_3^1)$$

$$x_1^2 = \frac{1}{8} \left( 21 - 2(-1.3) + 3\left(\frac{-5}{3}\right) \right)$$

$$x_1^2 = \frac{93}{40}$$

$$x_2^{1+1} = \frac{1}{a_{22}} (b_2 - a_{21}x_1^1 - a_{23}x_3^1)$$

$$x_2^{1+1} = \frac{1}{-10} (13 + x_1^1 - 4x_3^1)$$

$$x_2^2 = \frac{1}{-10} \left( 13 + \left(\frac{21}{8}\right) - 4\left(\frac{-5}{3}\right) \right)$$

$$x_2^2 = \frac{-107}{48}$$

$$x_3^{1+1} = \frac{1}{a_{33}} (b_3 - a_{31}x_1^1 - a_{32}x_2^1)$$

$$x_3^{m+1} = \frac{1}{9} (-15 - 2x_1^m - 3x_2^m)$$

$$x_3^{1+1} = \frac{1}{9} (-15 - 2x_1^1 - 3x_2^1)$$

$$x_3^2 = \frac{1}{9} \left( -15 - 2 \left( \frac{21}{8} \right) - 3(-1.3) \right)$$

$$x_3^2 = \frac{-109}{60}$$

**m = 2 (3<sup>rd</sup> iteration) :**

$$x_1^{2+1} = \frac{1}{a_{11}} (b_1 - a_{12}x_2^2 - a_{13}x_3^2)$$

$$x_1^{2+1} = \frac{1}{8} (21 - 2x_2^2 + 3x_3^2)$$

$$x_1^3 = \frac{1}{8} \left( 21 - 2 \left( \frac{-107}{48} \right) + 3 \left( \frac{-109}{60} \right) \right)$$

$$x_1^3 = \frac{2401}{960} \cong 2.501$$

$$x_2^{2+1} = \frac{1}{a_{22}} (b_2 - a_{21}x_1^2 - a_{23}x_3^2)$$



$$x_2^{2+1} = \frac{1}{-10} (13 + x_1^2 - 4x_3^2)$$

$$x_2^3 = \frac{1}{-10} \left( 13 + \left( \frac{93}{40} \right) - 4 \left( \frac{-109}{60} \right) \right)$$

$$x_2^3 = \frac{-2711}{1200} \cong -2.259$$

$$x_3^{2+1} = \frac{1}{a_{33}} (b_3 - a_{31}x_1^2 - a_{32}x_2^2)$$

$$x_3^{2+1} = \frac{1}{9} (-15 - 2x_1^2 - 3x_2^2)$$

$$x_3^3 = \frac{1}{9} \left( -15 - 2 \left( \frac{93}{40} \right) - 3 \left( \frac{-107}{48} \right) \right)$$

$$x_3^3 = \frac{-1037}{720} \cong -1.440$$

# Seidel Iterative Methods

## The Gauss Method

**The system given by :**

***Same like Jacobi but different in Simplifying :***

$$x_1^{m+1} = \frac{1}{a_{11}} (b_1 - a_{12}x_2^m - a_{13}x_3^m)$$

$$x_2^{m+1} = \frac{1}{a_{22}} (b_2 - a_{21}x_1^{m+1} - a_{23}x_3^m)$$

$$x_3^{m+1} = \frac{1}{a_{33}} (b_3 - a_{31}x_1^{m+1} - a_{32}x_2^{m+1})$$

**Example. Apply the Gauss method to solve :**

$$8x_1 + 2x_2 - 3x_3 = 21$$

$$-x_1 - 10x_2 + 4x_3 = 13$$

$$2x_1 + 3x_2 + 9x_3 = -15$$

***Simplifying three equations :***

$$x_1^{m+1} = \frac{1}{a_{11}} (b_1 - a_{12}x_2^m - a_{13}x_3^m)$$

$$x_1^{m+1} = \frac{1}{8} (21 - 2x_2^m + 3x_3^m)$$

$$x_2^{m+1} = \frac{1}{a_{22}} (b_2 - a_{21}x_1^{m+1} - a_{23}x_3^m)$$

$$x_2^{m+1} = \frac{1}{-10} (13 + x_1^{m+1} - 4x_3^m)$$

$$x_3^{m+1} = \frac{1}{a_{33}} (b_3 - a_{31}x_1^{m+1} - a_{32}x_2^{m+1})$$

$$x_3^{m+1} = \frac{1}{9} (-15 - 2x_1^{m+1} - 3x_2^{m+1})$$

$$(x_1^{(0)}, x_2^{(0)}, x_3^{(0)}, \dots, x_n^{(0)}) = 0$$

**Let m = 0 (1<sup>st</sup> iteration) :**

$$x_1^{m+1} = \frac{1}{a_{11}} (b_1 - a_{12}x_2^m - a_{13}x_3^m)$$

$$x_1^{0+1} = \frac{1}{8} (21 - 2x_2^0 + 3x_3^0)$$

$$x_1^1 = \frac{1}{8} (21)$$

$$x_1^1 = \frac{21}{8}$$

$$x_2^{m+1} = \frac{1}{a_{22}} (b_2 - a_{21}x_1^{m+1} - a_{23}x_3^m)$$

$$x_2^{0+1} = \frac{1}{-10} (13 + x_1^{0+1} - 4x_3^0)$$

$$x_2^1 = \frac{1}{-10} (13 + x_1^1)$$

$$x_2^1 = \frac{1}{-10} \left( 13 + \frac{21}{8} \right)$$

$$x_2^1 = \frac{-25}{16}$$

$$x_3^{m+1} = \frac{1}{a_{33}} (b_3 - a_{31}x_1^{m+1} - a_{32}x_2^{m+1})$$

$$x_3^{0+1} = \frac{1}{9} (-15 - 2x_1^{0+1} - 3x_2^{0+1})$$

$$x_3^1 = \frac{1}{9} (-15 - 2x_1^1 - 3x_2^1)$$

$$x_3^1 = \frac{1}{9} \left( -15 - 2 \left( \frac{21}{8} \right) - 3 \left( \frac{-25}{16} \right) \right)$$

$$x_3^1 = \frac{-83}{48}$$

**m = 1 (2<sup>nd</sup> iteration) :**

$$x_1^{m+1} = \frac{1}{a_{11}} (b_1 - a_{12}x_2^m - a_{13}x_3^m)$$

$$x_1^{1+1} = \frac{1}{8} (21 - 2x_2^1 + 3x_3^1)$$

$$x_1^2 = \frac{1}{8} \left( 21 - 2 \left( \frac{-25}{16} \right) + 3 \left( \frac{-83}{48} \right) \right)$$

$$x_1^2 = \frac{303}{128}$$

$$x_2^{m+1} = \frac{1}{a_{22}} (b_2 - a_{21}x_1^{m+1} - a_{23}x_3^m)$$

$$x_2^{1+1} = \frac{1}{-10} (13 + x_1^{1+1} - 4x_3^1)$$

$$x_2^2 = \frac{1}{-10} (13 + x_1^2 - 4x_3^1)$$

$$x_2^2 = \frac{1}{-10} \left( 13 + \frac{303}{128} - 4 \left( \frac{-83}{48} \right) \right)$$

$$x_2^2 = \frac{-8557}{3840}$$

$$x_3^{m+1} = \frac{1}{a_{33}} (b_3 - a_{31}x_1^{m+1} - a_{32}x_2^{m+1})$$

$$x_3^{1+1} = \frac{1}{9} (-15 - 2x_1^{1+1} - 3x_2^{1+1})$$

$$x_3^2 = \frac{1}{9} (-15 - 2x_1^2 - 3x_2^2)$$

$$x_3^2 = \frac{1}{9} \left( -15 - 2 \left( \frac{303}{128} \right) - 3 \left( \frac{-8557}{3840} \right) \right)$$

$$x_3^2 = -1.449$$

**m = 2 (3<sup>rd</sup> iteration) :**

$$x_1^{m+1} = \frac{1}{a_{11}} (b_1 - a_{12}x_2^m - a_{13}x_3^m)$$

$$x_1^{2+1} = \frac{1}{8} (21 - 2x_2^2 + 3x_3^2)$$

$$x_1^3 = \frac{1}{8} \left( 21 - 2 \left( \frac{-8557}{3840} \right) + 3(-1.449) \right)$$

$$x_1^3 = 2.638$$

$$x_2^{m+1} = \frac{1}{a_{22}} (b_2 - a_{21}x_1^{m+1} - a_{23}x_3^m)$$

$$x_2^{2+1} = \frac{1}{-10} (13 + x_1^{2+1} - 4x_3^2)$$

$$x_2^3 = \frac{1}{-10} (13 + x_1^3 - 4x_3^2)$$

$$x_2^3 = \frac{1}{-10} (13 + 2.638 - 4(-1.449))$$

$$x_2^3 = -2.143$$

$$x_3^{m+1} = \frac{1}{a_{33}} (b_3 - a_{31}x_1^{m+1} - a_{32}x_2^{m+1})$$

$$x_3^{2+1} = \frac{1}{9} (-15 - 2x_1^{2+1} - 3x_2^{2+1})$$

$$x_3^3 = \frac{1}{9} (-15 - 2x_1^3 - 3x_2^3)$$

$$x_3^3 = \frac{1}{9} (-15 - 2(2.638) - 3(-2.143))$$

$$x_3^3 = -1.538$$

# **Matrices**

## **The Cramer Method**

**The system given by :**

**The General Form**

**Suppose you have the following system of equations:**

$$a_1x_1 + b_1x_2 + c_1x_3 = d_1$$

$$a_2x_1 + b_2x_2 + c_2x_3 = d_2$$

$$a_3x_1 + b_3x_2 + c_3x_3 = d_3$$

**2. Calculate the Main Determinant (D( $\Delta$ ))**

**First, find the determinant of the coefficient matrix. Let's call it D:**

$$\Delta = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}$$



**Important: If  $D = 0$ , Cramer's Rule cannot be used because the system either has no solution or infinitely many solutions.**

### **3. Calculate Variable Determinants ( $\Delta x_1, \Delta x_2, \Delta x_3$ )**

**To find the determinant for a specific variable, replace that variable's column in the main matrix with the constants from the right side of the equations ( $d_1, d_2, d_3$ )**

- . For  $\Delta x_1$ : Replace the first column ( $x_1$  coefficients).**
- . For  $\Delta x_2$ : Replace the second column ( $x_2$  coefficients).**
- . For  $\Delta x_3$ : Replace the third column ( $x_3$  coefficients).**

#### **4. Solve for the Variables**

**The final step is to divide each variable determinant by the main determinant:**

$$x_1 = \frac{\Delta x_1}{\Delta}$$

$$x_2 = \frac{\Delta x_2}{\Delta}$$

$$x_3 = \frac{\Delta x_3}{\Delta}$$

**Example. Apply the Cramer method to solve :**

$$8x_1 + 2x_2 - 3x_3 = 21$$

$$-x_1 - 10x_2 + 4x_3 = 13$$

$$2x_1 + 3x_2 + 9x_3 = -15$$

$$\Delta = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}$$

$$\Delta = \begin{pmatrix} 8 & 2 & -3 \\ -1 & -10 & 4 \\ 2 & 3 & 9 \end{pmatrix}$$

$$\begin{array}{ccccc} 8 & \cdot & 2 & \cdot & -3 & \cdot & 8 & \cdot & 2 \\ -1 & \cdot & -10 & \cdot & 4 & \cdot & -1 & \cdot & -10 \\ 2 & \cdot & 3 & \cdot & 9 & \cdot & 2 & \cdot & 3 \end{array}$$

$$\begin{aligned} \Delta = & 8 * -10 * 9 + 2 * 4 * 2 - 3 * -1 * 3 - 2 \\ & * -1 * 9 - 8 * 4 * 3 + 3 * -10 * 2 \\ = & -833 \end{aligned}$$

$$\Delta x_1 = \begin{pmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{pmatrix}$$

$$\Delta x_1 = \begin{pmatrix} 21 & 2 & -3 \\ 13 & -10 & 4 \\ -15 & 3 & 9 \end{pmatrix}$$

$$\begin{array}{ccccc} 21 & \cdot & 2 & \cdot & -3 & \cdot & 21 & \cdot & 2 \\ 13 & \cdot & -10 & \cdot & 4 & \cdot & 13 & \cdot & -10 \\ \cdot & -15 & \cdot & 3 & \cdot & 9 & \cdot & -15 & \cdot & 3 \end{array}$$

$$\begin{aligned} \Delta x_1 = & 21 * -10 * 9 + 2 * 4 * -15 - 3 * 13 * 3 \\ & - 2 * 13 * 9 - 21 * 4 * 3 + 3 * -10 \\ & * -15 = -2163 \end{aligned}$$

$$\Delta x_2 = \begin{pmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{pmatrix}$$

$$\Delta x_2 = \begin{pmatrix} 8 & 21 & -3 \\ -1 & 13 & 4 \\ 2 & -15 & 9 \end{pmatrix}$$

$$\begin{array}{ccccc} 8 & \cdot & 21 & \cdot & -3 & \cdot & 8 & \cdot & 21 \\ -1 & \cdot & 13 & \cdot & 4 & \cdot & -1 & \cdot & 13 \\ \cdot & 2 & \cdot & -15 & \cdot & 9 & \cdot & 2 & \cdot & -15 \end{array}$$

$$\begin{aligned} \Delta x_2 &= 8 * 13 * 9 + 21 * 4 * 2 - 3 * -1 - 15 \\ &\quad - 21 * -1 * 9 - 8 * 4 * -15 + 3 * 13 * 2 \\ &= 1806 \end{aligned}$$

$$\Delta x_3 = \begin{pmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{pmatrix}$$

$$\Delta x_3 = \begin{pmatrix} 8 & 2 & 21 \\ -1 & -10 & 13 \\ 2 & 3 & -15 \end{pmatrix}$$

$$\begin{array}{ccccc} 8 & \cdot & 2 & \cdot & 21 \\ -1 & \cdot & -10 & \cdot & 13 \\ 2 & \cdot & 3 & \cdot & -15 \end{array}$$

$$\Delta x_3 = 8 * -10 * -15 + 2 * 13 * 2 + 21 * -1 * 3 - 2 * -1 * -15 - 8 * 13 * 3 - 21 * -10 * 2 = 1267$$

$$x_1 = \frac{\Delta x_1}{\Delta}$$

$$x_1 = \frac{-2163}{-833}$$

$$x_1 = 2.596$$

$$x_2 = \frac{\Delta x_2}{\Delta}$$

$$x_2 = \frac{1806}{-833}$$

$$x_2 = -2.168$$

$$x_3 = \frac{\Delta x_3}{\Delta}$$

$$x_3 = \frac{1267}{-833}$$

$$x_3 = -1.521$$

# **Matrices**

## **The Matrix Method**

**The system given by :**

**The General Form**

$$***AX = B***$$

***Where:***

***A: The Coefficient Matrix (contains the numbers next to  $x_1, x_2, x_3$ ).***

***X: The Variable Vector (contains  $x_1, x_2, x_3$  that we want to find).***

***B: The Constant Vector (contains the results on the right side of the equations).***

***The Solution Formula***

$$***X = A^{-1} \cdot B***$$



**Example. Apply the Matrix method to solve :**

$$8x_1 + 2x_2 - 3x_3 = 21$$

$$-x_1 - 10x_2 + 4x_3 = 13$$

$$2x_1 + 3x_2 + 9x_3 = -15$$

$$\left[ \begin{array}{ccc|ccc} 8 & 2 & -3 & \vdots & 1 & 0 & 0 \\ -1 & -10 & 4 & \vdots & 0 & 1 & 0 \\ 2 & 3 & 9 & \vdots & 0 & 0 & 1 \end{array} \right]$$

**A**

**I**

***Step 1: Normalize the first row ( $R_1$ )***

***Divide  $R_1$  by 8 to make the pivot element 1:***

$$\left[ \begin{array}{ccc|ccc} 1 & \frac{1}{4} & \frac{-3}{8} & \vdots & \frac{1}{8} & 0 & 0 \\ -1 & -10 & 4 & \vdots & 0 & 1 & 0 \\ 2 & 3 & 9 & \vdots & 0 & 0 & 1 \end{array} \right]$$

***Step 2: Eliminate elements below the first pivot***

$$R_1 + R_2 \rightarrow R_2$$

$$R_3 - 2R_1 \rightarrow R_3$$

$$\left[ \begin{array}{ccc|ccc} 1 & 0.25 & -0.375 & : & 0.125 & 0 & 0 \\ 0 & -9.75 & 3.625 & : & 0.125 & 1 & 0 \\ 0 & 2.5 & 9.75 & : & -0.25 & 0 & 1 \end{array} \right]$$

***Step 3: Normalize the second row ( $R_2$ )***

***Divide  $R_2$  by  $-9.75$ :***

$$\left[ \begin{array}{ccc|ccc} 1 & 0.25 & -0.375 & : & 0.125 & 0 & 0 \\ 0 & 1 & -\frac{29}{78} & : & \frac{-1}{78} & \frac{-4}{39} & 0 \\ 0 & 2.5 & 9.75 & : & -0.25 & 0 & 1 \end{array} \right]$$

***Step 4: Eliminate elements above and below the second pivot***

$$R_1 - 0.25R_2 \rightarrow R_1$$

$$R_3 - 2.5R_2 \rightarrow R_3$$

$$\left[ \begin{array}{cccccc} 1 & 0 & \frac{-11}{39} & \vdots & \frac{5}{39} & \frac{1}{39} & 0 \\ 0 & 1 & -\frac{29}{78} & \vdots & \frac{-1}{78} & \frac{-4}{39} & 0 \\ 0 & 0 & \frac{833}{78} & \vdots & -\frac{17}{78} & \frac{10}{39} & 1 \end{array} \right]$$

**Step 5: Normalize the third row ( $R_3$ )**

**Divide  $R_3$  by  $\frac{833}{78}$ :**

$$\left[ \begin{array}{cccccc} 1 & 0 & \frac{-11}{39} & \vdots & \frac{5}{39} & \frac{1}{39} & 0 \\ 0 & 1 & -\frac{29}{78} & \vdots & \frac{-1}{78} & \frac{-4}{39} & 0 \\ 0 & 0 & 1 & \vdots & -\frac{1}{49} & \frac{20}{833} & \frac{78}{833} \end{array} \right]$$

***Step 6: Eliminate elements above the third pivot***

$$R_1 + \frac{11}{39} R_3 \rightarrow R_1$$

$$R_2 + \frac{29}{78} R_3 \rightarrow R_2$$

$$\left[ \begin{array}{cccc|ccc} 1 & 0 & 0 & \vdots & \frac{6}{49} & \frac{27}{833} & \frac{22}{833} \\ 0 & 1 & 0 & \vdots & -\frac{1}{49} & -\frac{78}{833} & \frac{29}{833} \\ 0 & 0 & 1 & \vdots & -\frac{1}{49} & \frac{20}{833} & \frac{78}{833} \end{array} \right]$$

**I**

**$A^{-1}$**

$$A^{-1} = \left[ \begin{array}{ccc} \frac{6}{49} & \frac{27}{833} & \frac{22}{833} \\ -\frac{1}{49} & -\frac{78}{833} & \frac{29}{833} \\ -\frac{1}{49} & \frac{20}{833} & \frac{78}{833} \end{array} \right]$$

$$B = \begin{bmatrix} 21 \\ 13 \\ -15 \end{bmatrix}$$

$$X = A^{-1} \cdot B$$

$$X = \begin{bmatrix} \frac{102}{833} & \frac{27}{833} & \frac{22}{833} \\ -\frac{17}{833} & -\frac{78}{833} & \frac{29}{833} \\ \frac{17}{833} & \frac{20}{833} & \frac{78}{833} \end{bmatrix} * \begin{bmatrix} 21 \\ 13 \\ -15 \end{bmatrix}$$

$$X = \begin{bmatrix} 2.596 \\ -2.168 \\ -1.521 \end{bmatrix}$$

# **Error Analysis**

## **1. Definition**

**In numerical methods, an Error occurs due to the difference between the Exact Solution (obtained from direct methods like Cramer's Rule & Matrix's Rule) and the Approximate Solution (obtained from iterative methods like Jacobi Method & Gauss Method)**

## **2. Error Formulas**

**There are two main types of errors we calculate to measure the accuracy of our results:**

**Absolute Error ( $E_a$ ): It is the magnitude of the difference between the exact value and the approximate value.**

$$E_a = |x_{exact} - x_{approx}|$$

**Relative Error ( $E_r$ ):** It is the absolute error divided by the exact value, often expressed as a percentage to show the impact of the error relative to the size of the value.

$$E_r = \left| \frac{x_{exact} - x_{approx}}{x_{exact}} \right| * 100\%$$

#### **4. Example from your Results**

**Absolute Error:**

$$Exact\ x_1 = 2.596$$

$$Approx(Jacobi)\ x_1 = 2.501$$

$$Approx(Gauss)\ x_1 = 2.638$$

$$E_a = |x_{exact} - x_{approx}|$$

$$E_{a1}(Jacobi) = |2.596 - 2.501|$$

$$E_{a1}(Jacobi) = 0.095$$

$$E_a = |x_{exact} - x_{approx}|$$

$$E_{a1}(\text{Gauss}) = |2.596 - 2.638|$$

$$E_{a1}(\text{Gauss}) = 0.042$$

$$\text{Exact } x_2 = -2.168$$

$$\text{Approx}(\text{Jacobi}) x_2 = -2.259$$

$$\text{Approx}(\text{Gauss}) x_2 = -2.143$$

$$E_a = |x_{exact} - x_{approx}|$$

$$E_{a2}(\text{Jacobi}) = |-2.168 - -2.259|$$

$$E_{a2}(\text{Jacobi}) = 0.091$$

$$E_a = |x_{exact} - x_{approx}|$$

$$E_{a2}(\text{Gauss}) = |-2.168 - -2.143|$$

$$E_{a2}(\text{Gauss}) = 0.025$$

$$\text{Exact } x_3 = -1.521$$

$$\text{Approx}(\text{Jacobi}) x_3 = -1.440$$

$$\text{Approx}(\text{Gauss}) x_3 = -1.538$$

$$E_a = |x_{exact} - x_{approx}|$$

$$E_{a3}(\text{Jacobi}) = |-1.521 - -1.440|$$



$$E_{a3}(\text{Jacobi}) = 0.081$$

$$E_{a3} = |x_{exact} - x_{approx}|$$

$$E_{a3}(\text{Gauss}) = |-1.521 - -1.538|$$

$$E_{a3}(\text{Gauss}) = 0.017$$

### **Relative Error:**

$$E_r = \left| \frac{x_{exact} - x_{approx}}{x_{exact}} \right| * 100\%$$

$$E_{r1}(\text{Jacobi}) = \left| \frac{2.596 - 2.501}{2.596} \right| * 100$$

$$E_{r1}(\text{Jacobi}) = 3.659\%$$

$$E_{r1}(\text{Gauss}) = \left| \frac{2.596 - 2.638}{2.596} \right| * 100$$

$$E_{r1}(\text{Gauss}) = 1.617\%$$

$$E_r = \left| \frac{x_{exact} - x_{approx}}{x_{exact}} \right| * 100\%$$

$$E_{r2}(\text{Jacobi}) = \left| \frac{-2.168 - -2.259}{-2.168} \right| * 100$$

$$E_{r2}(\text{Jacobi}) = 4.197\%$$

$$E_{r2}(\text{Gauss}) = \left| \frac{-2.168 - -2.143}{-2.168} \right| * 100$$

$$E_{r2}(\text{Gauss}) = 1.153\%$$

$$E_r = \left| \frac{x_{exact} - x_{approx}}{x_{exact}} \right| * 100\%$$

$$E_{r3}(\text{Jacobi}) = \left| \frac{-1.521 - -1.440}{-1.521} \right| * 100$$

$$E_{r3}(\text{Jacobi}) = 5.325\%$$

$$E_{r2}(\text{Gauss}) = \left| \frac{-1.521 - -1.538}{-1.521} \right| * 100$$

$$E_{r3}(\text{Gauss}) = 1.117\%$$